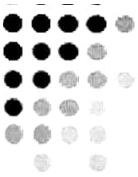


8.1 INTRODUCTION

The metals handbook defines heat treatment as “a combination of heating and cooling operations timed and applied to a metal or alloy in the solid state to produce desired properties”. Eventhough the heat treatment is applicable to many metallic materials, the present discussion is confined to steels only.

Steels are classified generally based on their composition, as plain carbon steels and alloy steels. The alloy steels contain significant levels of alloying elements other than carbon. Plain carbon steels are further classified into low carbon steels (0.1 to 0.3% C), mild steel or medium carbon steel (0.3% to 0.6 % C) and high carbon steel (above 0.6 %). The heat treatment can bring changes in the microstructure and hence the properties can be modified as per requirements.

HEAT TREATMENTS



- Definition:
 - ***a combination of heating and cooling operations timed and and applied to a metal or alloy in the solid state to produce desired properties***

Objectives/ Purposes of heat treatment



- To improve mechanical properties
- To improve surface hardness
- To increase resistance to wear, corrosion etc
- To improve machinability
- To relieve internal stresses
- To refine grain structure after hot working
- To improve magnetic and electrical properties

Stages of heat treatment : Fig. 7.9 illustrate the principle of heat treatment in which the structure and therefore the properties of steels can be changed. The three stages of heat treatment are

1. Heating to required temperature
2. Holding at this temperature for a period of time to attain uniform temperature throughout the section. This stage is called soaking.
3. Cooling the steel at specified rate.

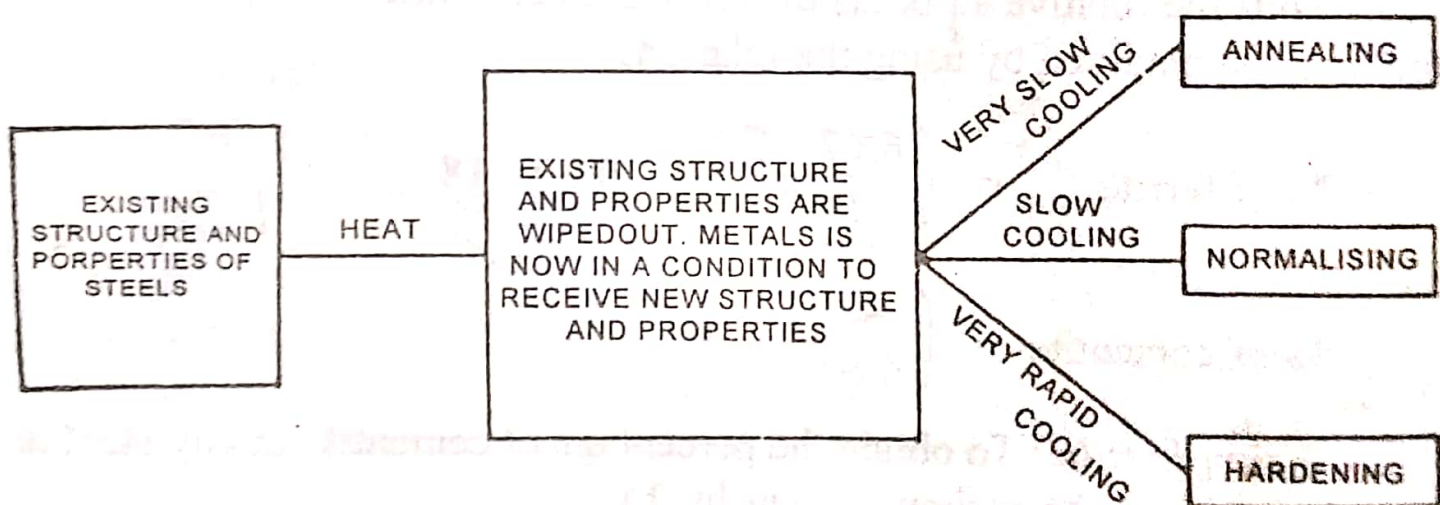
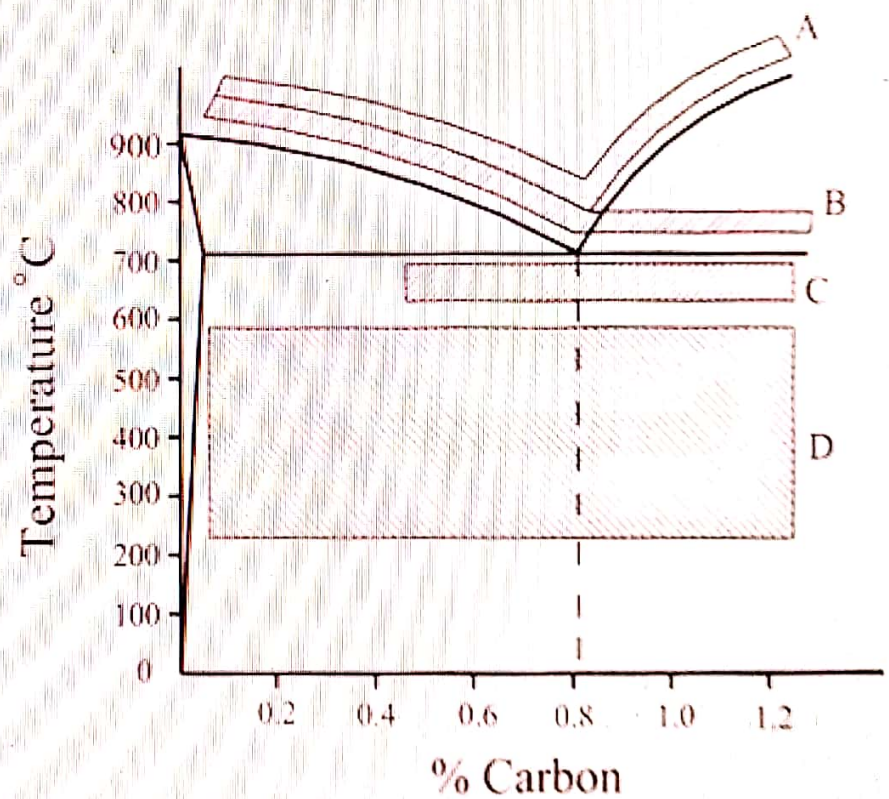


Fig. 7.9 PRINCIPLE OF HEAT TREATMENT

Heat Treatments

- A – Normalising
- B – Annealing or Hardening
- C – Spheroidising or Process Annealing
- D - Tempering



13.14 Heat Treatment

The process of heat treatment is carried out first by heating the metal and then cooling it in the caustic soda solution, brine, water, oil or air. The purpose of heat treatment is to soften the metal, to change the grain size, to modify the structure of the material and to relieve the stresses set up in the material after hot or cold working. The various heat treatment processes commonly employed in engineering practice are as follows:

1. **Annealing.** It is one of the most important process of heat treatment of steel. Following are four types of annealing:

- (a) **Full annealing.** The purpose of full annealing is to soften the metal, to refine the grain structure, to relieve the stresses and to remove trapped gases in the metal. The process consists of heating the steel $30^{\circ}\text{C} - 50^{\circ}\text{C}$ above the upper critical temperature for hypo-eutectoid steel and by the same temperature above the lower critical temperature for hyper-eutectoid steels. It is held at this temperature for sometime and then cooled slowly in the furnace.
- (b) **Process annealing.** It is also known as low temperature annealing or sub-critical annealing. This process is used for relieving the internal stresses previously set up in the metal and for increasing the machinability of the steel. In this process, steel is heated to a temperature below or close to the lower critical temperature (generally $550^{\circ}\text{C} - 650^{\circ}\text{C}$), held at this temperature for sometime and then cooled slowly.
- (c) **Spheroidise annealing (spheroidising).** It is usually applied to high carbon tool steels which are difficult to machine. The operation consists of heating the steel to a

temperature slightly above the lower critical temperature (730°C to 770°C). It is held at this temperature for sometime and then cooled slowly to a temperature of 600°C . The spheroidising improves the machinability of steels, but lowers the hardness and tensile strength.

- (d) *Diffusion annealing (Homogenization)*. This process is mainly used for ingots and large castings. After diffusion annealing, the castings undergo full annealing to improve their properties or to refine grain structure. The process consists of heating the steel to a high temperature (1100°C – 1200°C). It is held at this temperature for 8 to 20 hours and then cooled to 800°C – 850°C inside the furnace for a period of about 6 to 8 hours. It is further cooled in the air to room temperature.
2. *Normalising*. The normalising is done for the following purposes:
- (a) To refine the grain structure of the steel to improve machinability, tensile strength and structure of weld.
 - (b) To remove strains caused by cold working processes.
 - (c) To remove dislocations caused in the internal structure of the steel due to hot working.
 - (d) To improve certain mechanical and electrical properties.

The process of normalising consists of heating the steel 30°C – 50°C above its upper critical temperature for hypo-eutectoid steels or A_{cm} line for hyper-eutectoid steels. It is held at this temperature for about fifteen minutes and then allowed to cool down in still air. The process of normalising is frequently applied to castings and forgings etc.

3. *Hardening*. The main objects of hardening are

- (a) To increase the hardness of the metal so that it can resist wear.
- (b) To enable it to cut other metals, *i.e.* to make it suitable for cutting tools.

The process of hardening consists of heating the metal to a temperature of 30°C to 50°C above the upper critical point for hypo-eutectoid steels and by the same temperature above the lower critical temperature for hyper-eutectoid steels. It is held at this temperature for a considerable time and then quenched (cooled suddenly) in a suitable cooling medium.

6. *Tempering*. The tempering (also known as drawing) is done for the following reasons:
- (a) To reduce brittleness of the hardened steel and thus to increase ductility.
 - (b) To remove internal stresses caused by rapid cooling of steel.
 - (c) To make steel tough to resist shock and fatigue.

The tempering process consists of reheating the hardened steel to some temperature below the lower critical temperature, followed by any desired rate of cooling.

7. *Surface hardening* or *Case hardening*. In many engineering applications, it is desirable that a steel being used should have a hardened surface to resist wear and tear. At the same time, it should have soft and tough interior or core so that it is able to absorb any shocks etc. This type of

treatment is applied to gears, ball bearings, railway wheels etc. The various surface or case hardening processes are as follows:

- (a) Carburising ; (b) Cyaniding ; (c) Nitriding; (d) Induction hardening ; and
- (e) Flame hardening.

- **Carburising** – introducing carbon in to the surface of steel. In this process low carbon steel is heated in contact with carbonaceous material from which steel absorbs carbon
- **Nitriding** – introducing nitrogen in to the surface of steels by heating and holding it at a suitable temp. In contact with partially dissociated ammonia or suitable medium
- **Cyaniding** – both carbon and nitrogen are added to the surface layers of steel

Flame hardening

- Gas flames raise the temperature of the outer surface above the upper critical temp.
- Water jets quench the component and thereby hardens it

Induction hardening

- Induced eddy currents heat the surface of the steel very quickly and is quickly followed by jets of water to quench the component.
- A hard outer layer is created with a soft core.
- The slideways on a lathe are induction hardened